PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

ARTICLE DETAILS

TITLE (PROVISIONAL)	Effects of Baduanjin Exercise on Motor Function, Balance and Gait in Parkinson's Disease: A Systematic Review and Meta-analysis
AUTHORS	Lai, Jinghui; Cai, Yangfan; Yang, Liyan; Xia, Min; Cheng, Xi; Chen, Ying

VERSION 1 – REVIEW

REVIEWER	Tsukita, Kazuto Kyoto University, Department of Neurology
REVIEW RETURNED	07-Sep-2022

GENERAL COMMENTS	Lai et al conducted a systematic review and meta-analysis of the effect of Baduanjin exercise on motor and balance/gait function in patients with Parkinson's disease (PD). This meta-analysis comes from the aggregation of a few studies of very small sample sizes and heterogeneity in the intervention methodology (i.e. Baduanjin exercise) certainly make the interpretation of this meta-analysis difficult; however, considering the growing interest in the exercise intervention and PD, this meta-analysis is worth to be published to fuel future reserch of Baduanjin exercise in patients with PD. Generally, the meta-analysis is well-executed and I have only some minor points. 1) Please provide the funnel plot because publication bias greatly influences the outcome measure in this kind of meta-analysis. 2) Please provide the measure of adherence because it is the most important thing in exercise RCTs. It is known that even when patients are instructed to perform high-intensity exercise four times per week, the average number of times per week was three, according to a recent landmark study, SPARXII (JAMA neurol 2018). It will be very interesting to know how well patients adhere to the original protocol in the case of the Badwanjin exercise. 3) If possible, please describe where exercise interventions (home, hospital, studio, etc.) were undertook. 4) Please provide the sex information in the Table 1. 5) Are there any outcomes relating to the non-motor symptoms? If any, please provide it.

REVIEWER	McKee , Kathleen Intermountain Healthcare
REVIEW RETURNED	19-Sep-2022

GENERAL COMMENTS	Lai et al present the largest meta-analysis of the effects of Baduanjin on PD motor function to-date. They ultimately include ten randomized controlled trials and conclude that Baduanjin significantly improves multiple PD outcome measures (UPDRS, TUG, BBS, etc.). They should be applauded for raising awareness of Baduanjin as a possible treatment for PD. However their conclusions reach beyond what their analysis supports. The heterogeneity of the studies included and use of “medication treatment” as an acceptable control group calls into question the conclusion they reach. In RCTs with “medication treatment” as a control group one can not exclude the possibility of a ‘placebo’ effect on the intervention group – simply increased attention and social interaction could be the result of the improvement on outcome measures. Additionally ‘balance board’ is used along sinde Baduanjin in quite a few of the studies in the intervention group – the analysis needs to address how this is or is not dis-entangled in interpreting the results. The paper would also benefit from greater explination of Baduanjin as many readers of BMJ Open are likely not familiar with it. In particular – how does it compare to quigong or is it a sub-form of quigong. Also with the description of Baduanjin provided I do not see how it qualifies as a moderate-aerobic activity. The authors also need to contextualize their study with the other literature reported on Baduanjin – particularly the 2017 meta-analysis by Zho etal (Zou L, SasaKi JE, Wang H, Xiao Z, Fang Q, Zhang M. A Systematic Review and Meta-Analysis Baduanjin Qigong for Health Benefits: Randomized Controlled Trials. Evid Based Complement Alternat Med. 2017;2017:4548706) and the recently published meta-analysis by Hao et al (Hao Z, Zhang X, Chen P. Effects of Ten Different Exercise Interventions on Motor Function in Parkinson's Disease Patients-A Network Meta-Analysis of Randomized Controlled Trials. Brain Sci. 2022 May 27;12(6):698.) – which did not find a robust benfit from Baduanjin – although it only included three studies and to be fair it also did not find robust benefit to treadmill training in direct contradiction to SPARX2 study. The discussion would be better spent on this contextualization than on far reaching speculation about mechanism and possible disease modifying effects of Baduanjin. The introduction also needs to be tightened significantly and there are several stylistic edits that need to be made. • EG: First sentence could be edited as follows to read more smoothly: Parkinson's disease (PD) represents the second most common neurodegenerative disease, which is characterized by the decrease OF dopaminergic neurons in the substantia nigra pars compacta (SNpc) 1. According to the global survey of nervous system diseases, the incidence and prevalence of PD. [There are additional edits very similar to these that need to be made throughout.] • Full stop should occur prior to endnote • Tenses shift throughout manuscript • Page 5 line 10 makes no sense • Page 7 line 31: at should be “a” • Page 10, line 46 – paragraph starts by talking about 8 months but shifts at end to talk about weeks

	• Page 13 line 24: I don't' understand "movement disorders" in this line

VERSION 1 – AUTHOR RESPONSE

Reviewer: 1
Dr. Kazuto Tsukita, Kyoto University
Comments to the Author:
Lai et al conducted a systematic review and meta-analysis of the effect of Baduanjin exercise on motor and balance/gait function in patients with Parkinson's disease (PD). This meta-analysis comes from the aggregation of a few studies of very small sample sizes and heterogeneity in the intervention methodology (i.e. Baduanjin exercise) certainly make the interpretation of this meta-analysis difficult; however, considering the growing interest in the exercise intervention and PD, this meta-analysis is worth to be published to fuel future reserch of Baduanjin exercise in patients with PD. Generally, the meta-analysis is well-executed and I have only some minor points.

1 Please provide the funnel plot because publication bias greatly influences the outcome measure in this kind of meta-analysis.
Response: We thank the reviewer for the suggestion. The manuscript was modified, accordingly.

2. Please provide the measure of adherence because it is the most important thing in exercise RCTs. It is known that even when patients are instructed to perform high-intensity exercise four times per week, the average number of times per week was three, according to a recent landmark study, SPARXII (JAMA neurol 2018). It will be very interesting to know how well patients adhere to the original protocol in the case of the Badwanjin exercise.
Response: We thank the reviewer for the careful review. The appropriate information was added to address this issue in the revised manuscript.

3. If possible, please describe where exercise interventions (home, hospital, studio, etc.) were undertook.
Response: We thank the reviewer for the careful review. The appropriate information was added in the revised manuscript.

4. Please provide the sex information in the Table 1.
Response: We thank the reviewer for the careful review. This information was added in the revised manuscript.

5. Are there any outcomes relating to the non-motor symptoms? If any, please provide it.
Response: We thank the reviewer for the careful review. All studies included in the review evaluated the effect of Baduanjin on the motor symptoms of patients, and did not involve non-motor symptoms.

Reviewer: 2
Dr. Kathleen McKee , Intermountain Healthcare
Comments to the Author:
Lai et al present the largest meta-analysis of the effects of Baduanjin on PD motor function to-date. They ultimately include ten randomized controlled trials and conclude that Baduanjin significantly improves multiple PD outcome measures (UPDRS, TUG, BBS, etc.). They should be applauded for raising awareness of Baduanjin as a possible treatment for PD.

1.However their conclusions reach beyond what their analysis supports. The heterogeneity of the studies included and use of "medication treatment" as an acceptable control group calls into question

the conclusion they reach. In RCTs with “medication treatment” as a control group one can not exclude the possibility of a ‘placebo’ effect on the intervention group – simply increased attention and social interaction could be the result of the improvement on outcome measures. Additionally ‘balance board’ is used along sinde Baduanjin in quite a few of the studies in the intervention group – the analysis needs to address how this is or is not dis-entangled in interpreting the results.
Response: We thank the reviewer for the comment. For all original studies included in the review, the control group received other treatments (e.g. drug therapy, balance board training, conventional treatment, and conventional physical therapy), while the experimental group performed Baduanjin exercises in addition to the intervention used in the control group. Therefore, the studies included for the analysis are relatively uniform, and the meta-analysis results are reliable.

2. The paper would also benefit from greater explination of Baduanjin as many readers of BMJ Open are likely not familiar with it. In particular – how does it compare to quigong or is it a sub-form of quigong. Also with the description of Baduanjin provided I do not see how it qualifies as a moderate-aerobic activity.
Response: We thank the reviewer for the suggestion. Baduanjin is a traditional medical and health care practice in ancient China. It is a type of Qigong (Reference: General Administration of Sports of China fitness Qigong management center. Two Hundred Questions of Fitness Qigong [M]. Beijing: People's Sports Publishing House 2007: 3). Metabolic equivalent (MET) is one of the most important measures of exercise intensity. In general, activities with <3 MET values are considered as low-intensity activities, activities with 3-6 MET values are considered as moderate-intensity activities, and activities with 6-9 MET values are considered as high-intensity activities. Baduanjin energy consumption ranges with 3-5 MET, according to the exercise intensity measurement standard for moderate-intensity activity. This information was added in the revised manuscript.

3. The authors also need to contextualize their study with the other literature reported on Baduanjin – particularly the 2017 meta-analysis by Zho etal (Zou L, SasaKi JE, Wang H, Xiao Z, Fang Q, Zhang M. A Systematic Review and Meta-Analysis Baduanjin Qigong for Health Benefits: Randomized Controlled Trials. Evid Based Complement Alternat Med. 2017;2017:4548706) and the recently published meta-analysis by Hao et al (Hao Z, Zhang X, Chen P. Effects of Ten Different Exercise Interventions on Motor Function in Parkinson's Disease Patients-A Network Meta-Analysis of Randomized Controlled Trials. Brain Sci. 2022 May 27;12(6):698.) – which did not find a robust benfit from Baduanjin – although it only included three studies and to be fair it also did not find robust benefit to treadmill training in direct contradiction to SPARX2 study. The discussion would be better spent on this contextualization than on far reaching speculation about mechanism and possible disease modifying effects of Baduanjin.
Response: We thank the reviewer for the suggestion. The appropriate information was added to address this issue in the revised manuscript.

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Response: We thank the reviewer for the suggestion. The manuscript was modified, accordingly.

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